

Kevin's Mathematical Olympiad (Round 1)

Mock Practice Paper made by Kevin Hu - 20/03/2026

EXAM INSTRUCTIONS

- **Time allowed:** $3\frac{1}{2}$ hours.
- Each question is worth 10 marks. **Maximum mark:** 60
- Full written explanations and theoretical supporting arguments are required for each problem. Although figures may sometimes help, they are alone not sufficient.
- A correct answer with no explanation may earn very few marks; a nearly complete proof with a minor error may receive nearly full marks. Note that this paper will be marked using a $0^+/10^-$ scheme.
- **Equipment:** No calculators or protractors are permitted. Ruler and compasses may be used.

1. Find all pairs of prime numbers (p, q) such that

$$p^q + q^p$$

is also a prime number.

2. Let M be the set of points (x, y) in the Cartesian plane such that $x \in \{1, 2, 3, 4, 5\}$ and $y \in \{1, 2, 3, 4\}$. Suppose that 11 of these points are colored red. Prove that there must exist four red points which form the vertices of a rectangle with sides parallel to the coordinate axes.

3. Let m and n be positive integers. Prove that the sum

$$5^n + 5^m$$

can be expressed as the sum of two perfect squares **if and only if** $n - m$ is even.

4. Let $\triangle ABC$ be an acute-angled triangle such that $AB > AC$. Let M be the midpoint of the side BC . A point P is chosen inside the triangle AMC such that

$$\angle BAM = \angle PAC.$$

Let O_1, O_2 , and O_3 denote the circumcenters of $\triangle ABC, \triangle ABP$, and $\triangle ACP$, respectively.

Prove that the line AO_3 bisects the segment O_1O_2 .

5. Let x, y , and z be distinct non-negative real numbers. Prove that

$$\sum_{\text{cyc}} \frac{1}{(x-y)^2} \geq \frac{4}{xy + yz + zx}.$$

6. Let Γ denote the circumcircle of $\triangle ABC$. Let L_1 be the tangent to Γ at point C . Let L_2 be the line passing through B parallel to AC . Let D be the intersection of L_1 and L_2 .

Let L_3 be the tangent to Γ at point B . Let L_4 be the line passing through C parallel to AB . Let E be the intersection of L_3 and L_4 .

Let L be the intersection of BE and CD .

Let Γ_1 be the circumcircle of $\triangle BDC$. Let T be the intersection of Γ_1 and line AC such that $T \neq C$.

Let Γ_2 be the circumcircle of $\triangle BEC$. Let S be the intersection of Γ_2 and line AB such that $S \neq B$.

Prove that the ST , BC , and AL are concurrent.

— END OF PAPER —